HANNAH ARNOLD

First Class BSc (Hons) Artificial Intelligence and Robotics Graduate
Stoke-on-Trent, Staffordshire | Hannah.marie.arnold.hma@gmail.com | +447377151443
LinkedIn | Portfolio

PROFESSIONAL PROFILE

First Class BSc (Hons) Artificial Intelligence and Robotics graduate with practical experience developing embedded firmware in C on ARM-based STM32 microcontrollers. Experienced in integrating sensors, motors and peripheral hardware using GPIO, timers and I²C, alongside hardware-software integration, circuit testing and systematic debugging.

Strong interest in embedded software engineering, real-time systems and microcontroller-based development, with additional training in embedded C, STM32 development, hardware interfacing, communication protocols, debugging and real-time embedded systems. Comfortable working from technical datasheets and hardware documentation to investigate faults and develop reliable solutions. Keen to develop further expertise in high-integrity, secure and quality-focused embedded software engineering.

TECHNICAL SKILLS

Languages	C, C++, Python, HTML, G
Hardware & Communication	STM32, GPIO, Timers, I²C, Sensors, Motors, Circuit Testing, MyRio
Software Development	Git, GitHub, Visual Studio, LabVIEW, Portfolio website
Methods	Algorithm Design, Debugging, Hardware-Software Integration
Professional Strengths	Problem-solving, Organisation, Communication, Teamwork & collaboration, Leadership, Adaptability, Working effectively under pressure

TECHNICAL PROJECTS

Final Year Project: Sensor-Based Embedded System for Roller Coaster Ride Path Analysis | October 2025 - May 2026

Designed and developed an embedded system using an ARM-based STM32 microcontroller to collect and process rider and environmental data. Developed embedded firmware in C, interfacing sensors, motors, a camera module and other hardware peripherals through GPIO, timers and I²C. Implemented a state-machine control system to process sensor data and control connected hardware. Performed systematic hardware and firmware debugging, investigating faults across both software and electronic circuitry to improve system reliability. Regularly consulted technical datasheets and hardware manuals to determine component specifications, pin configurations, communication requirements and implementation constraints. Tested the integrated system iteratively to identify and resolve hardware/software integration issues.

Ambient Light Home Automation System | September 2026

Developed an STM32-based embedded control project representing a smart-home application. The system uses an ambient-light sensor to monitor surrounding light levels, adjusts LED brightness using PWM, and controls a motor to simulate automated window blinds in low-light conditions. Focused on sensor interfacing, GPIO, motor control and real-time decision-making.

Personal Technical Portfolio | May 2026 - Present

Portfolio: xxhansterxx.github.io

Developing and maintaining a personal technical portfolio documenting ongoing learning, embedded systems projects and independent technical development. The portfolio includes project documentation, technical experimentation and evidence of continued development beyond university coursework.

EDUCATION

The University Of Staffordshire | September 2023 - May 2026

BSc (Hons) Artificial Intelligence and Robotics - First Class Honours

Relevant areas included embedded systems, robotic programming, artificial intelligence, software development and electronics. Practical work included STM32 microcontrollers, C/C++, Python, GPIO, timers, sensors, I²C communication and hardware-software integration.

City of Stoke-on-Trent Sixth Form College | September 2021 - July 2023
A-Level studies included Mathematics, Physics, Further Mathematics and Extended Project Qualification (EPQ).

ADDITIONAL TRAINING

Communication Protocols: UART, USART, I²C, I³C, SPI, GPIO | Udemy | September 2026
Theory based training covering embedded communication protocols including UART/USART, I²C, I³C, SPI, GPIO with protocol signals, packet formats, communication modes, addressing, SPI clock modes and GPIO configuration.

Embedded Software Development with C Specialization | Educba | August 2026
Completed training focused on embedded C programming, STM32 microcontrollers, software development, hardware interfacing, debugging and real-time embedded systems.

Qt Academy – QML For Beginners | Qt Academy | August 2026
Completed a structured learning path covering QML syntax, Qt Quick, responsive interface development, Model-View-Delegate architecture, animations and transitions. Developed practical experience building and debugging QML applications through hands-on coding challenges, with emphasis on maintainable application development and Qt best practices..

Learn PLC and HMI from Scratch (Basic) Specialization | Packt | August 2026
Completed training in industrial automation covering Siemens S7-1200 PLC programming, TIA Portal, ladder logic, HMI design, system simulation and troubleshooting, with further study of SCL programming, PID control and industrial control systems.

EMPLOYMENT EXPERIENCE

Retail Assistant, Primark | April 2022 - Present
Working in a fast-paced retail environment has developed my customer service, communication and teamwork skills. I am trusted to work across multiple departments and operate till systems accurately, while adapting to changing priorities during busy trading periods.
This role has strengthened my reliability, attention to detail and ability to remain composed under pressure. These transferable skills support my transition into a professional technical environment where collaboration, accountability and consistent performance are essential.

LEADERSHIP AND WIDER EXPERIENCE

STEM Workshop Assistant | December 2025 – January 2026
Supported across 12 technical workshops in December introducing primary-school pupils to multiple areas in tech covering digital safety, cryptography, games, computer hardware and network media. In January, helped deliver a full-day visit for 44 pupils across two classes, leading an interactive session on programming robotic arms to carry out practical tasks.

Committee Member and Captain, Macclesfield Warriors Hockey Club | December 2024 - Present
As Team Captain and Committee Member, I contribute to team organisation, communication and fixture coordination. The role requires calm decision-making, leadership under pressure and the ability to support others while maintaining a positive and professional team environment.

Engineering Society, The University Of Staffordshire | September 2023 - May 2024
Participated in practical engineering workshops involving electronics assembly, soldering and prototype development. Activities also included CAD-supported 3D-printing and laser-cutting projects, supporting broader practical engineering awareness.

DRIVING LICENSE

Full UK Driving License